

Mridul Sehgal

✉ mridul.sehgalwork@gmail.com | 📞 8347758913
🌐 github.com/MS-Pro-2001 | 🏠 leetcode

PROFESSIONAL SUMMARY

Full-stack Software Engineer with experience building scalable web applications. Skilled in RESTful API design, microservices architecture, and clean code practices. Proficient in Agile workflows, consistently delivering optimized performance and high-quality user experiences. Strong problem-solving and collaboration skills demonstrated through successful project delivery.

SKILLS

Languages: C/C++, JavaScript, TypeScript, Python

Web: React.js, Next.js, Node.js, Express.js, Redux, HTML5, CSS3

Backend/Database: MongoDB, MySQL, PostgreSQL, Firebase, REST API, GraphQL

DevOps/Tools: Git, GitHub, AWS

Others: JWT, OAuth, Zustand, Google Earth Engine

WORK EXPERIENCE

Webosmotic, Surat

Jan 2024 - Present

Software Engineer

- Architected and developed a **full-stack**, admin dashboard for a community collaboration platform using **React**, **Redux Toolkit**, and a modular codebase reduced development time for new features by **35%**.
- Designed and integrated secure, scalable microservices-based **RESTful APIs**, implementing **role-based access control (RBAC)** to reduce unauthorized access incidents by **90%**.
- Created a reusable **React** component library achieving **95%** design fidelity, improving UI development speed by **40%** and ensuring consistent branding across the platform.
- Contributed to a cabinet e-commerce platform (**React**, **Fastify**, **Node.js**, **PostgreSQL**) supporting both admin and user roles, implemented an end-to-end **3D model viewer**, increasing user engagement by **50%**.
- Led **QA testing** and managed **deployment cycles** for the e-commerce application, reducing post-release issues by **60%**.
- Developed a **Next.js** and **MongoDB** application focusing on **accessibility** (**screen-reader support**, **keyboard navigation**, **WCAG-compliant color contrast**) to create an inclusive user experience.
- Implemented **internationalization (i18n)** for multi-language support and managed the full **CI/CD pipeline** (**Vercel**, **Netlify**) for seamless cloud deployment.
- Integrated **Jest** and **React Testing Library** to implement unit and integration tests for critical frontend components, improving code reliability and reducing regression bugs by **55%**.

SAC ISRO, Ahmedabad

May 2023 - July 2023

Data Scientist-Intern

- Automated the processing of over **10 years** of MODIS and GIMMS satellite datasets using **Google Earth Engine**, enabling scalable environmental analysis and reducing manual workload by **60%**.
- Built a **regression-based predictive model** to assess long-term vegetation trends, incorporating first and second-order (double derivative) analysis for enhanced trend accuracy.
- Visualized spatial-temporal vegetation changes through interactive maps and **time-series plots**, facilitating data-driven environmental insights.

EDUCATION

NIT, Surat

July 2019 - June 2024

Master of Science in Mathematics (CGPA: 8.61/10)

HOBBIES

- Table Tennis:** Passionate about fast-paced rallies and improving reflex-based gameplay.
- Cricket:** Enjoy playing both casual and competitive matches, especially batting and field strategy.